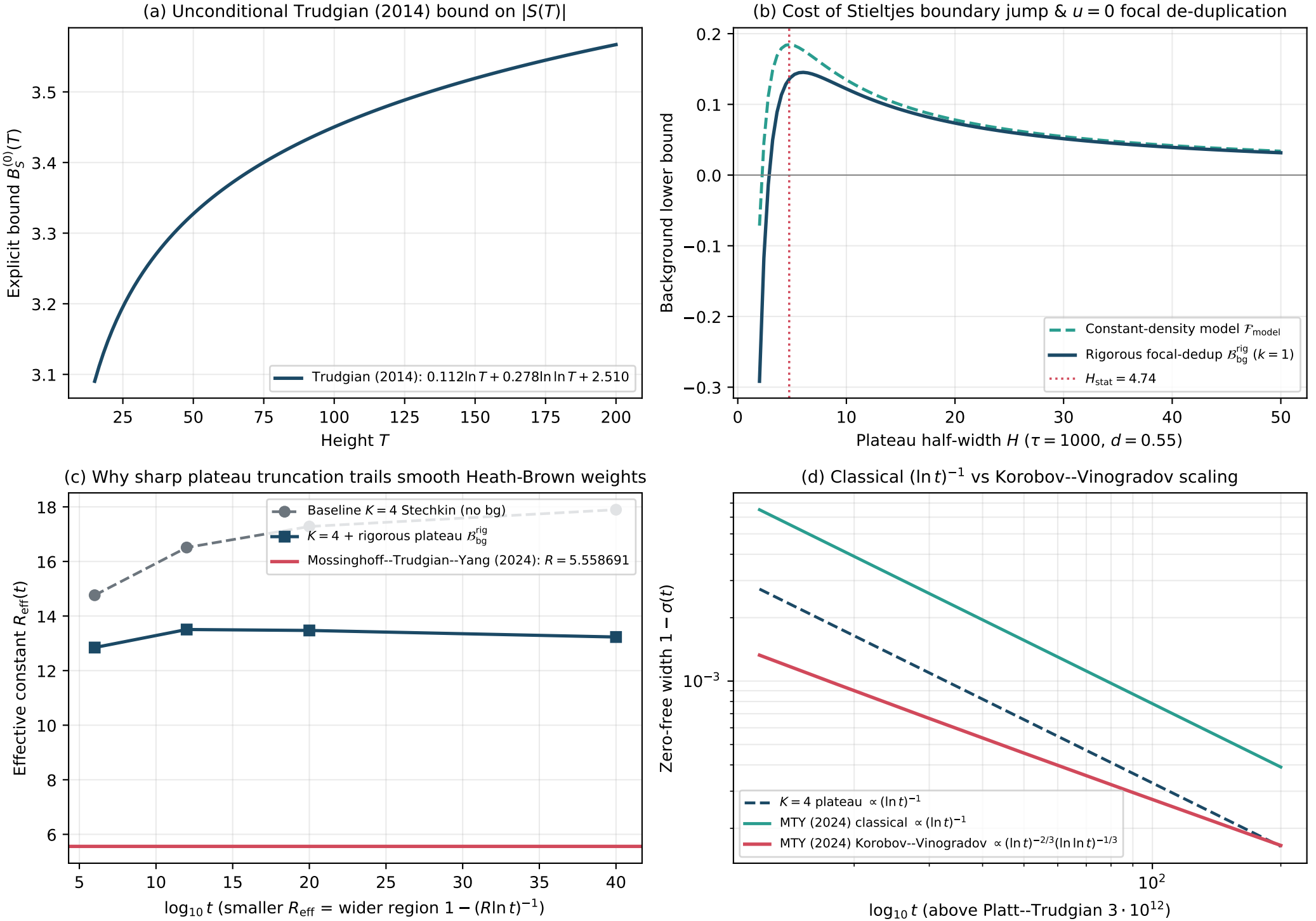
(a) Unconditional Trudgian (2014) bound on $|S(T)|$
Explicit bound $B_S^{(0)}(T)$
Height T
Trudgian (2014): $0.112\ln T + 0.278\ln\ln T + 2.510$
(b) Cost of Stieltjes boundary jump & $u = 0$ focal de-duplication
Background lower bound
Plateau half-width H ($\tau = 1000$, $d = 0.55$)
Constant-density model $\mathcal{F}_{\text{model}}$
Rigorous focal-dedup $\mathcal{B}_{\text{bg}}^{\text{rig}}$ ($k = 1$)
$H_{\text{stat}} = 4.74$
(c) Why sharp plateau truncation trails smooth Heath-Brown weights
Effective constant $R_{\text{eff}}(t)$
$\log_{10} t$ (smaller R_{eff} = wider region $1 - (R\ln t)^{-1}$)
Baseline $K = 4$ Stechkin (no bg)
$K = 4$ + rigorous plateau $\mathcal{B}_{\text{bg}}^{\text{rig}}$
Mossinghoff--Trudgian--Yang (2024): $R = 5.558691$
(d) Classical $(\ln t)^{-1}$ vs Korobov--Vinogradov scaling
Zero-free width $1 - \sigma(t)$
$\log_{10} t$ (above Platt--Trudgian $3 \cdot 10^{12}$)
$K = 4$ plateau $\propto (\ln t)^{-1}$
MTY (2024) classical $\propto (\ln t)^{-1}$
MTY (2024) Korobov--Vinogradov $\propto (\ln t)^{-2/3}(\ln\ln t)^{-1/3}$